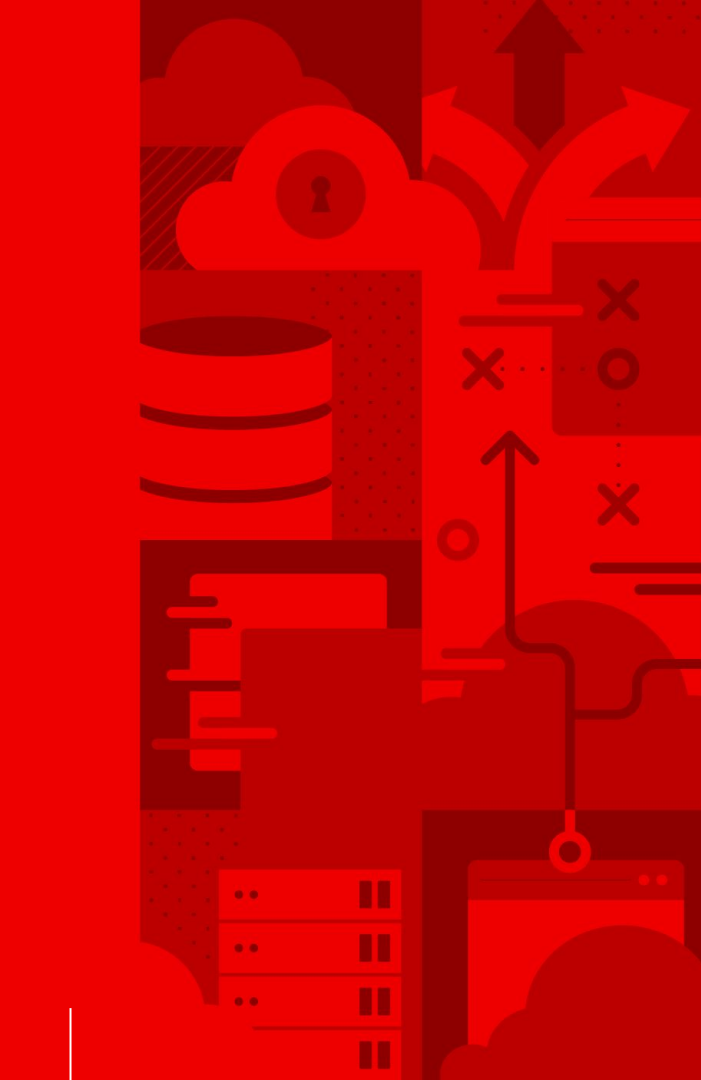


The fundamentals of technical writing

Tooling

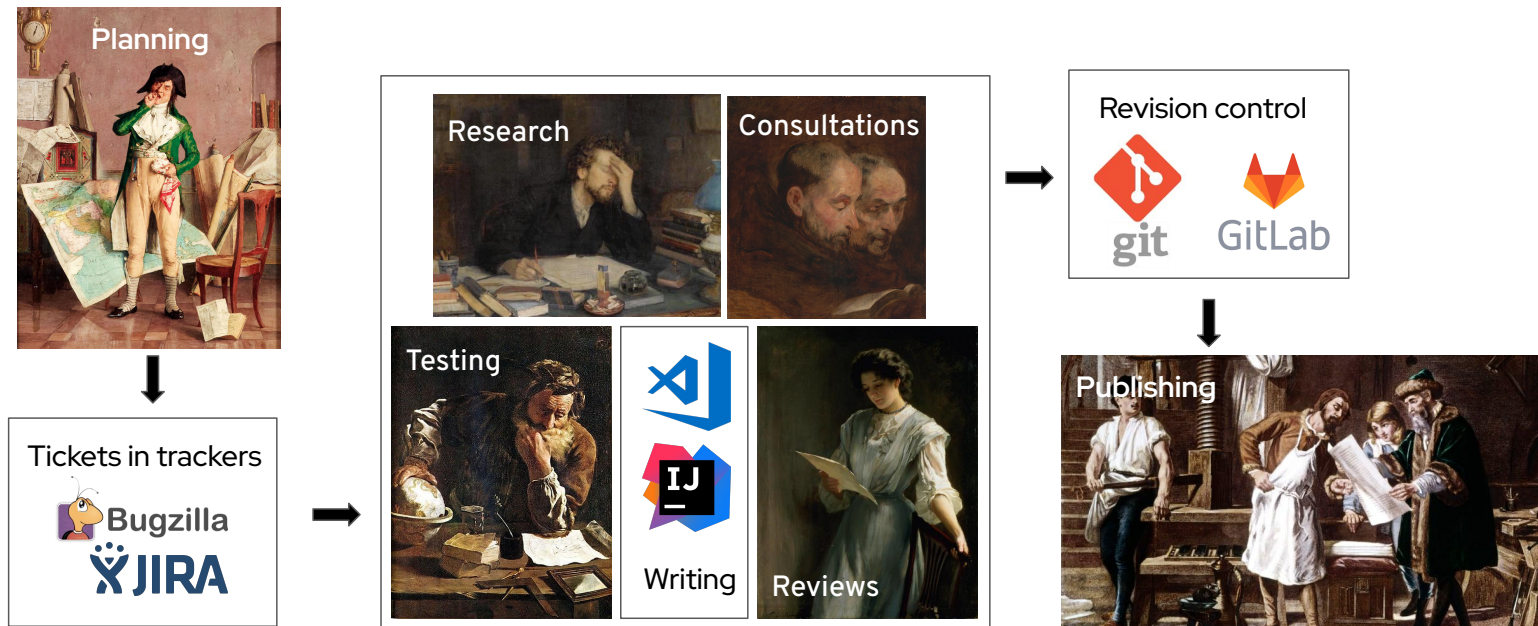
What we'll discuss today

- Introduction
- Editors and word processors
- Languages
- Collaboration tools
- Version control tools (Git)
- Graphics



Introduction

Reminder: How documentation actually happens



Credits:

- The Passion of Creation by Leonid Pasternak; wikimedia Commons, PD-Russia-expired, edited
- Domenico Fetti - Archimedes, Wikimedia Commons, PD-US, edited
- Anonymous, a Copy after a painting traditionally attributed to Van Dyck of Two Monks Reading, Wikimedia Commons, PD-US, edited
- Thomas Kennington - Reading the letter, Wikimedia Commons, PD-US, edited
- Giovanni Batista Quadroni - The Cartographer, Wikimedia Commons, PD-US, edited
- ClassicStock



Text editors,
Word processors

Text editors vs word processors

Editors (IDEs):

- vi/vim
- Visual Studio Code
- IntelliJ
- emacs
- etc.

Processors:

- MS Word
- LibreOffice
- Google Docs
- etc.

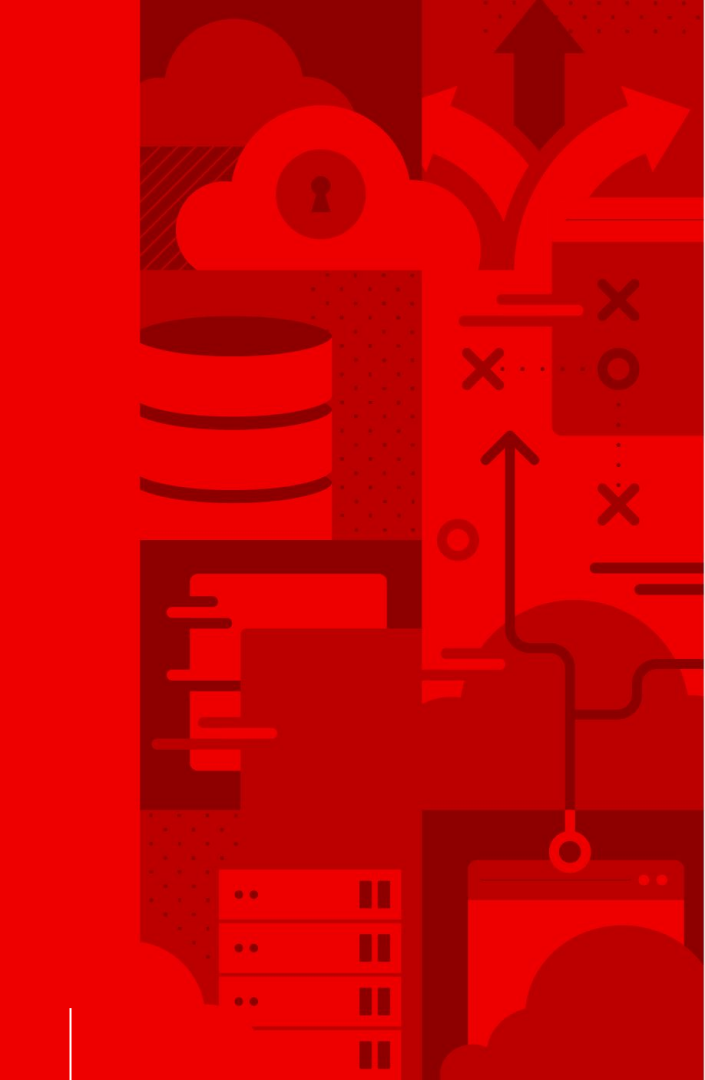
- Ultimately, choose whatever is comfortable and suitable for the project.

WYSIWYG

- Word processor (MS Word, LibreOffice Writer, Google Docs...)
- Granular control of all visual properties of all content elements
- Difficult to standardize, export, and version control
- Best for smaller doc sets with limited contributors

Docs as code

- Markup language (AsciiDoc, DocBook XML, Markdown...) + Text editor/IDE (Vim, VS Code, IntelliJ...)
- Determines types of content elements -> Visual properties rendered through stylesheets
- Easier standardization, exporting, and version control
- Best for larger doc sets with more contributors



Languages

What is a markup language?

“A **markup language** is a text-encoding system which specifies the structure and formatting of a document and potentially the relationships among its parts.”

- Wikipedia

XML-based languages

- DocBook XML, DITA XML, ...
- More granular semantic control
- Popular for commercial & corporate use
- Example:

```
<section>
  <title>Section name</title>
  <para>Intro sentence here.</para>

  <itemizedlist>
    <listitem>
      <para>First list entry</para>
    </listitem>
    <listitem>
      <para>Second list entry</para>
    </listitem>
  </itemizedlist>

</section>
```

Lightweight markups

- Markdown, AsciiDoc, RST, ...
- More human-readable
- Easy adoption -> popular in open source communities
- Example:

```
== Section name

Intro sentence here.

* First list entry
* Second list entry
```

Exercise: reverse-engineer this text

Get to know AsciiDoc

AsciiDoc is a plain text language used for documentation and books.

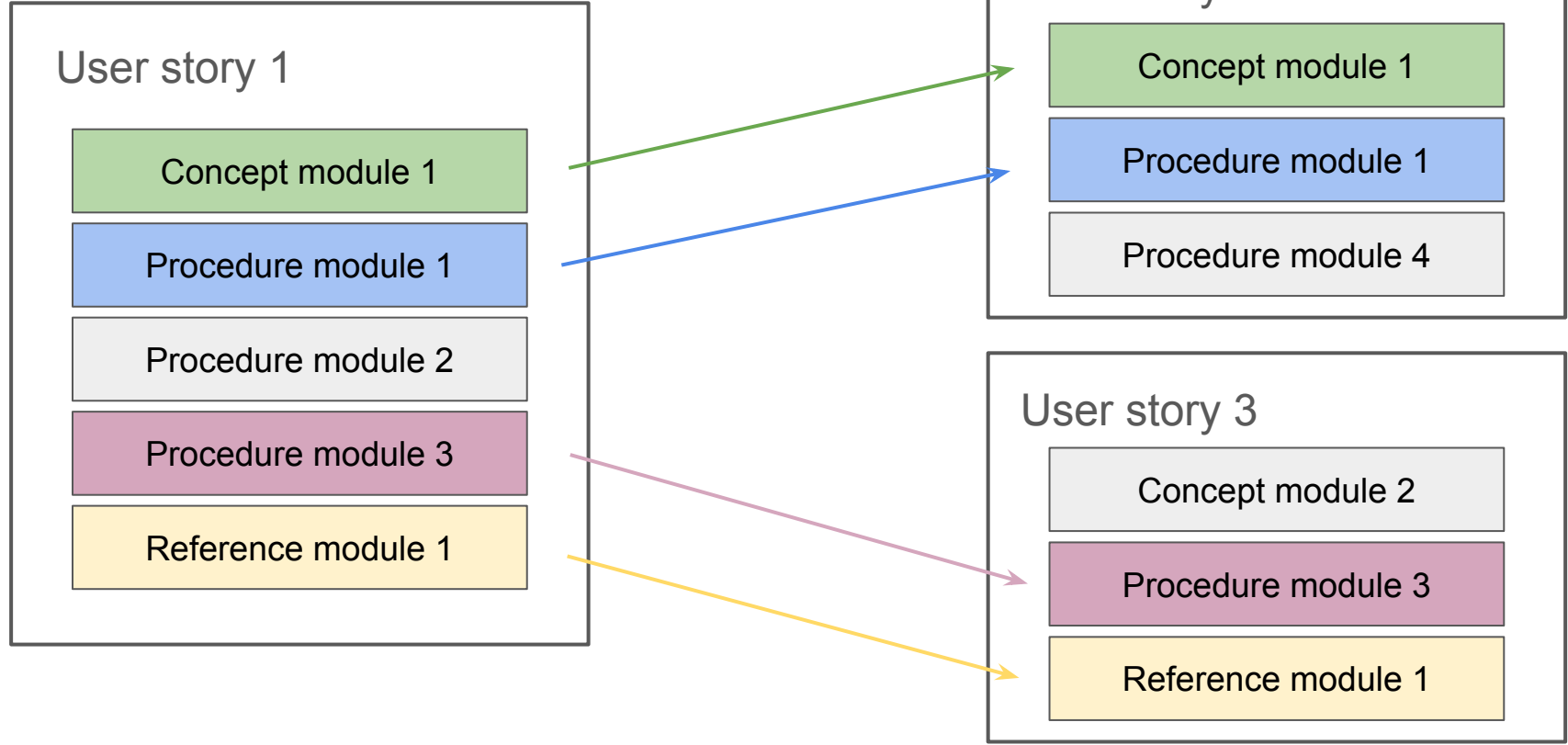
It's easy to start with but if you need help, go to [AsciiDoc Language Documentation](#).

Your next task

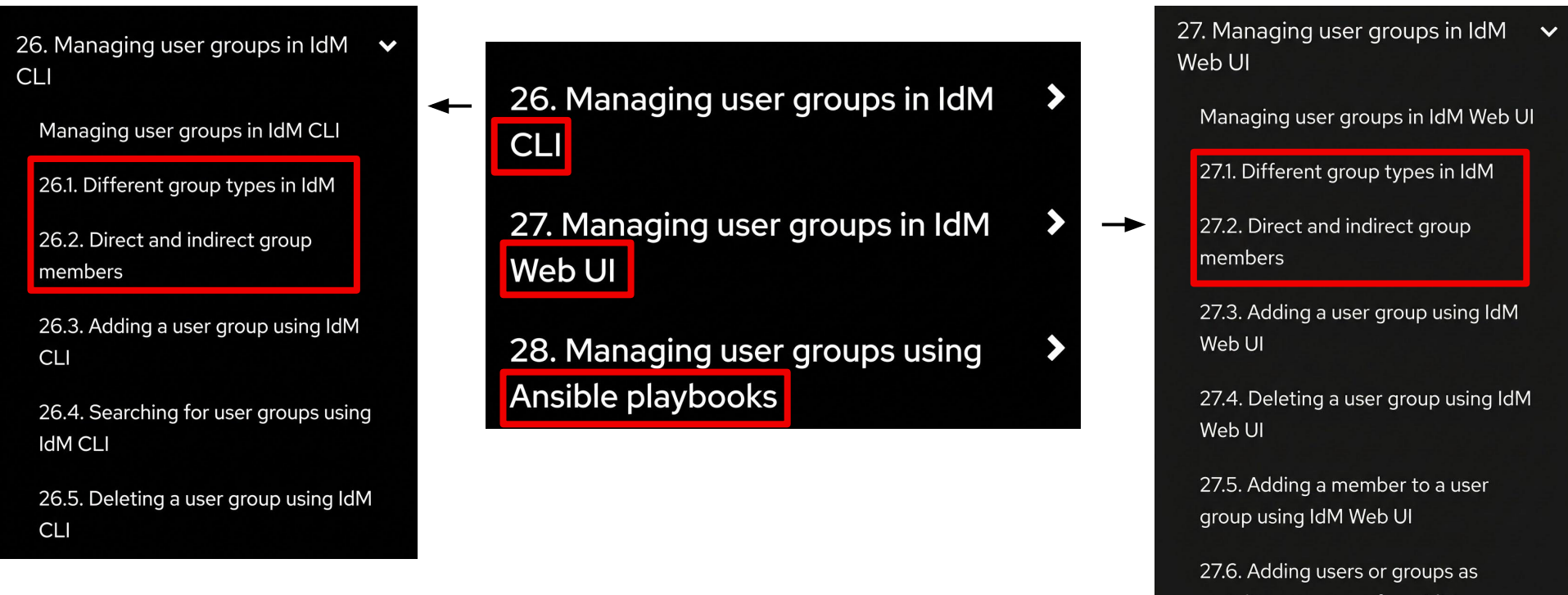
1. Open your browser and navigate to <https://asciidoc.org/#try>.
2. Rewrite this text.
3. Try different syntax elements.
 - For example a code block:

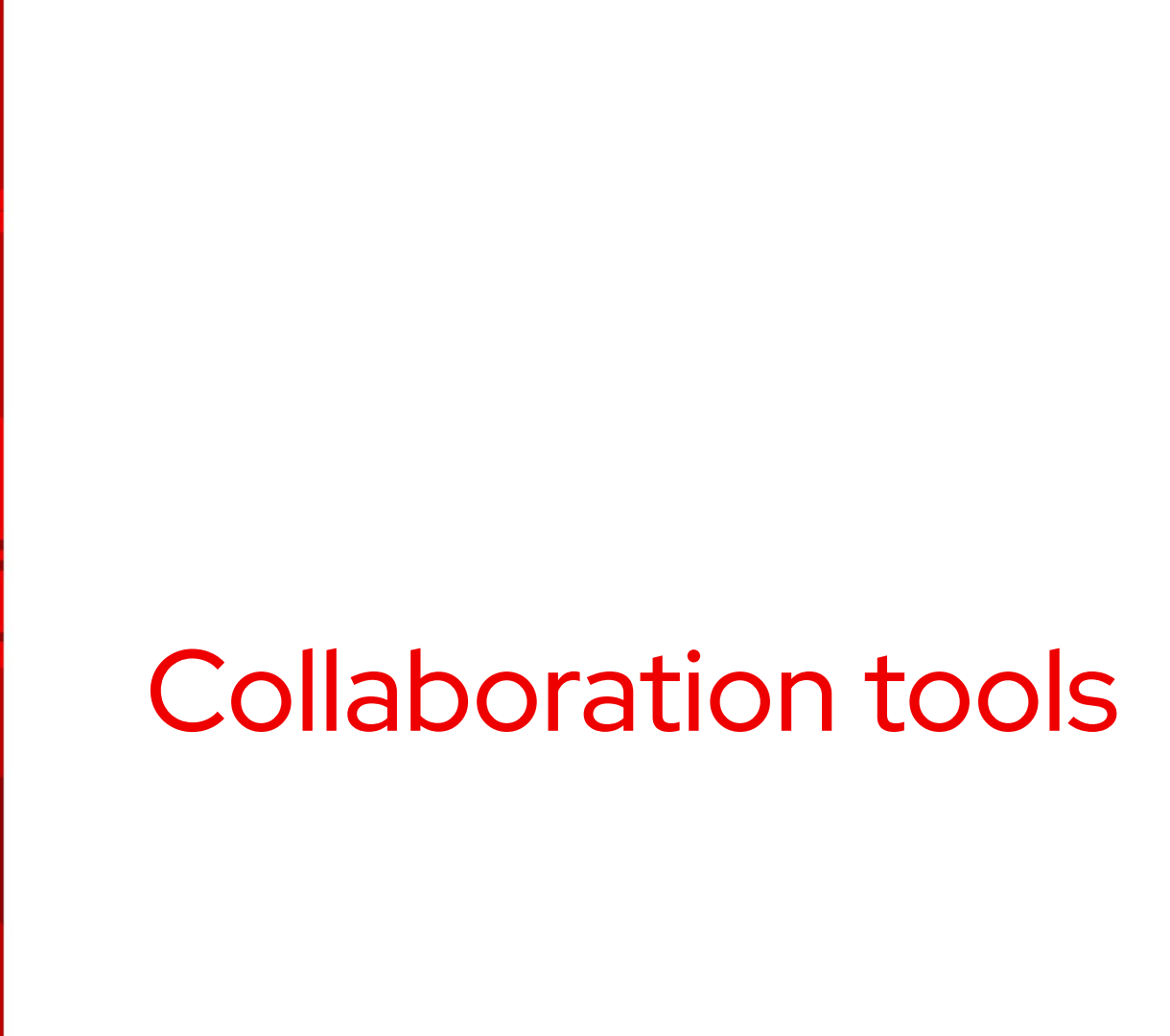
```
puts 'Good luck!'
```

Modular documentation



Single-sourcing in practice





Collaboration tools

Collaboration

- Communication (Slack, IRC, Google Meet, Zoom)



- Whiteboards (Miro)



- Project and task management tools (Jira, Confluence)



Bugzilla

- Version control (GitHub, GitLab, Bitbucket)



GitLab



on control

Version control

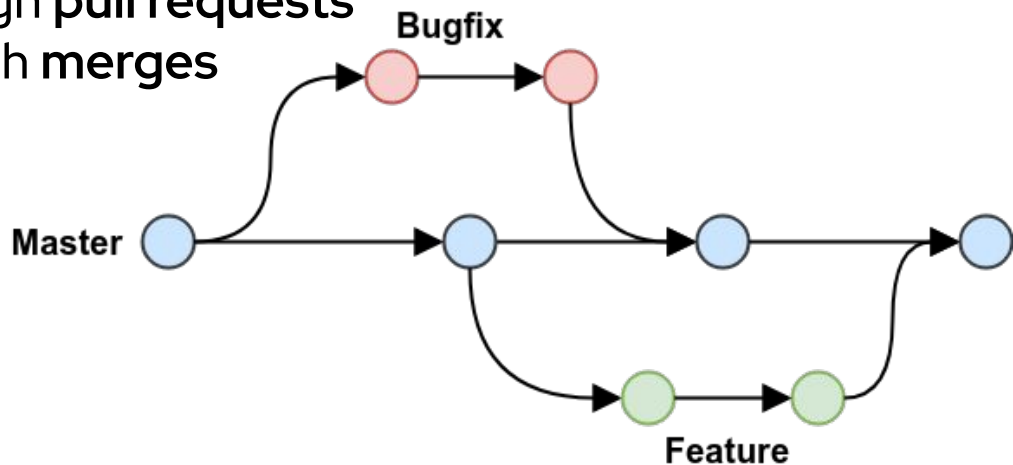
- Tracking and management of content changes
- Makes software development faster and more efficient
- Every modification is tracked in a special database
- Makes fixing issues easier
- The most widely used version control system is Git

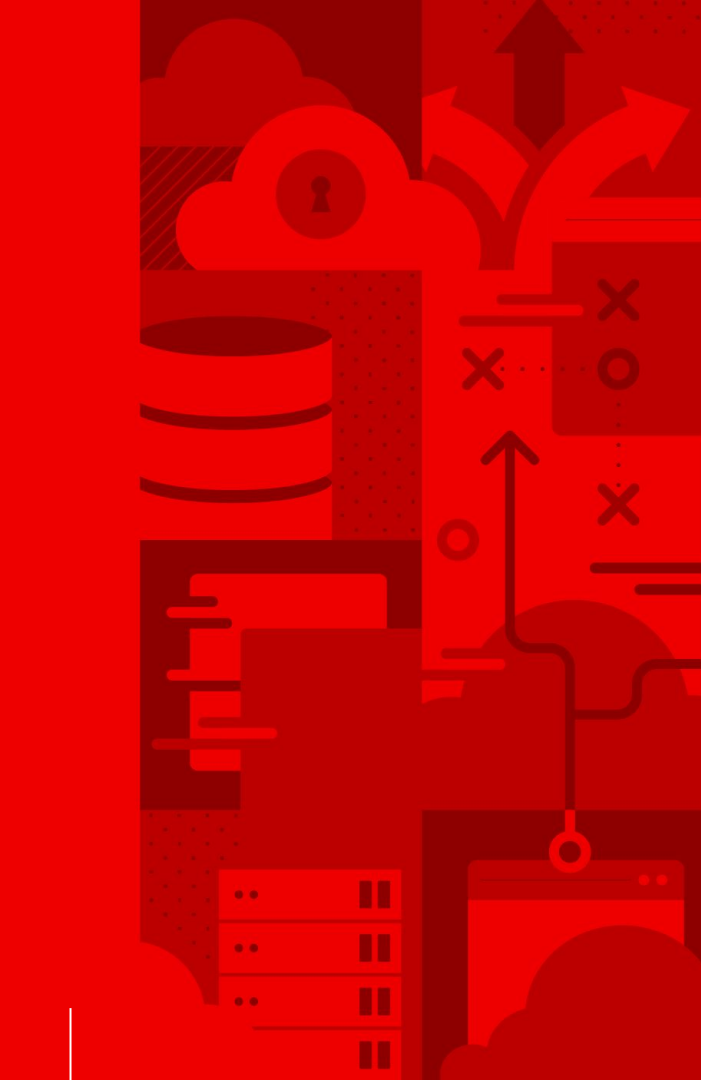


Git

For granular version control of software code (and docs as code)

- Code for projects resides in **repositories**
- Changes to code through **commits**
- Applying sets of changes through **branches**
- Review of branches through **pull requests**
- Combine branches through **merges**
- ...and much more



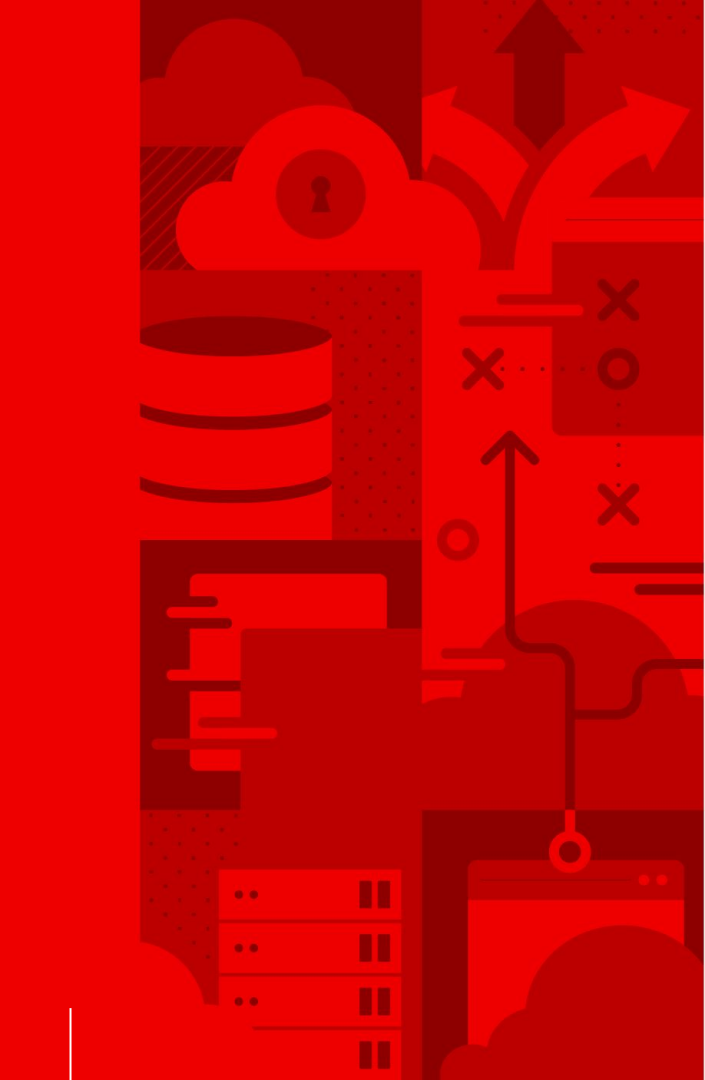


Tools for graphics

Graphics

- Screenshot capture
- Image manipulation (GIMP)
- Diagrams (Inkscape, diagrams.net, mermaid.js)

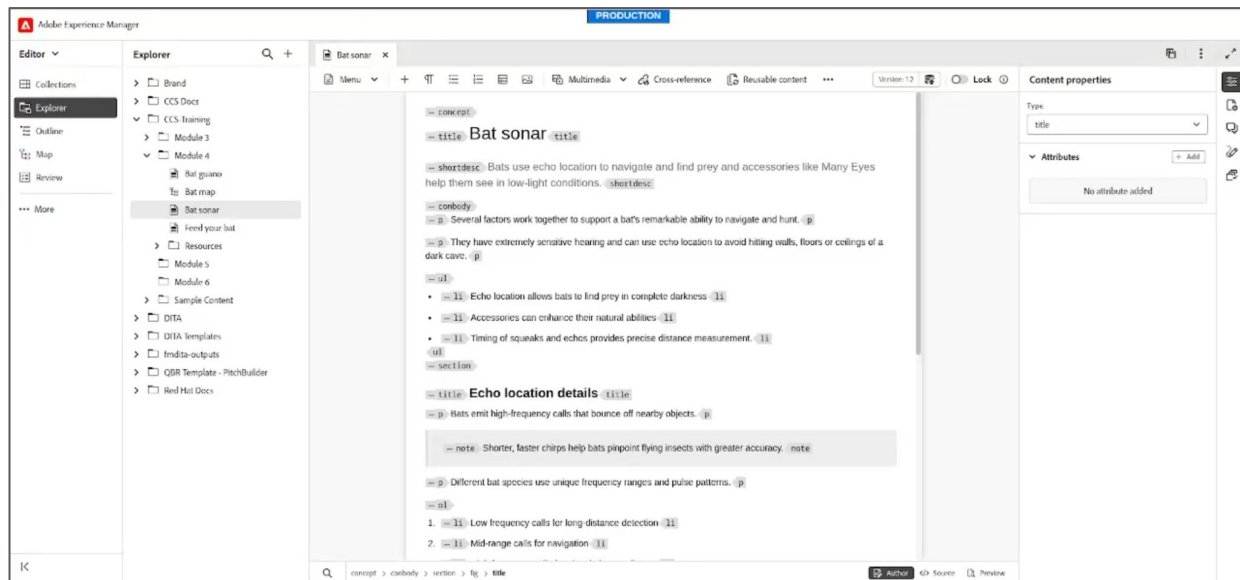




Content management systems (CMS)

Content management systems

- WordPress, Adobe Experience Manager, Drupal,...
- Combine multiple functions (Markup, WYSIWYG, version control, task tracking, publishing...)



Takeaways

- Tech writers generally use:
 - Text editors or CMSs to create and edit docs in a markup language (= docs as code)
 - Version control systems, such as Git, to track and manage changes to the docs source
 - Modular structure for single-sourcing docs content
 - Ticket trackers for managing workload
 - Graphics tools to work with screenshots and diagrams
 - Communication and teamwork tools
 - AI Tools (more info in the afternoon :-)



Exercise

Collaboration in Git – part 1

- **Task:** Add *a page about yourself* to the [wiki section of the course GitHub repository](#)
 - Share anything you feel like sharing (hobbies, favourite band, etc.)
 - No need to use real names or biographic data – keep in mind the repo is public
 - Use the GitHub web GUI and AsciiDoc markup.
- Time allotment: ~15 mins

Thank you